

Decision and Risk Analysis

Rational Choice under Uncertainty

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Risk vs. Uncertainty

Often a distinction is made between **risk** vs. **uncertainty**:

- A choice under **risk** is made when the outcomes are random and the probabilities of outcomes are known.
- A choice under **uncertainty** is made when the outcome probabilities are unknown.

Decision Making under Uncertainty: Example I

- You prepare to leave home, you wonder whether it will rain.
- Combining a quick assessment of the cloudy sky (visual data) with knowledge of weather patterns in your area, you estimate the probability of rain at 30%.
- You need to make a decision: should you carry an umbrella or not?

Decision Making under Uncertainty: Example I

		World State	
		Rain prob. = 0.3	No rain prob. = 0.7
Action	Do not take umbrella	wet & cold cost = 90	dry & happy cost = 0
	Take umbrella	slightly damp cost = 10	dry & burdened cost = 5

		World State	
		Rain prob. = 0.3	No rain prob. = 0.7
Action	Do not take umbrella	wet & cold cost = 40	dry & happy cost = 0
	Take umbrella	slightly damp cost = 20	dry & burdened cost = 10

Figure 1: (A) Walking in the rain without an umbrella results in a wet-and-cold outcome loss (cost) of 90; walking in the rain with an umbrella leaves one only slightly damp (loss (cost) of 10), etc.; (B) Another individual assigns different losses (costs)

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- Comparing the expected reward under alternative courses of action for individual (A) we obtain:

$$\begin{aligned}\mathbb{E}[r(s, a = \text{umbrella})] &= \mathbb{P}(s = \text{rain})(-10) + \mathbb{P}(s = \text{no rain})(-5) \\ &> \\ \mathbb{E}[r(s, a = \text{no umbrella})] &= \mathbb{P}(s = \text{rain})(-90) + \mathbb{P}(s = \text{no rain})0 \\ -10 \cdot 0.3 - 5 \cdot 0.7 &= -6.5 > -90 \cdot 0.3 + 0 \cdot 0.7 = -27\end{aligned}$$

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- For individual (B), the optimal action is to leave the umbrella at home: $-20 \cdot 0.3 - 10 \cdot 0.7 = -13 < -40 \cdot 0.3 + 0 \cdot 0.7 = -12$

Decision Making under Risk: Example II

- A company owns a tract of land that may contain oil. A geologist has determined with a chance of 1 in 4, there is oil in the tract.
- Another party has offered to purchase the land for \$90,000
- If oil is found (after drilling at a cost of \$100,000) the company's net profit is expected to be \$700,000
- If the land is dry then company would have incurred the drilling cost.

Decision Making under Risk: Example II

- The elements of this problem are summarized as follows:

Action	State of the World	
	Oil	Dry
Drill for oil	\$700	−\$100
Sell the land	\$90	\$90

Decision Making under Risk: Example II

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Action	State of the World	
	Oil	Dry
Drill for oil	\$700	−\$100
Sell the land	\$90	\$90

- What do we mean when we say a certain course of action is “optimal”?

Decision Making under Risk: Example II

- We are asked to compare two actions (or “lotteries”):

$$r(s, a = \text{drill}) = \begin{cases} \$700 & s = \text{Oil} \\ -\$100 & s = \text{Dry} \end{cases}$$

$$r(s, a = \text{sell}) = \begin{cases} \$90 & s = \text{Oil} \\ \$90 & s = \text{Dry} \end{cases}$$

- Choose the action with the higher **expected reward**.

Expected Reward Calculation

Let $\mathbb{P}(\text{Oil}) = p$ and $\mathbb{P}(\text{Dry}) = 1 - p$.

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Expected Value of Each Action

$$\mathbb{E}[r(\text{drill})] = p(700) + (1 - p)(-100) = 800p - 100$$

$$\mathbb{E}[r(\text{sell})] = 90$$

- The action with higher expected reward should be chosen.

Threshold Probability

We find the point where both actions are equally good:

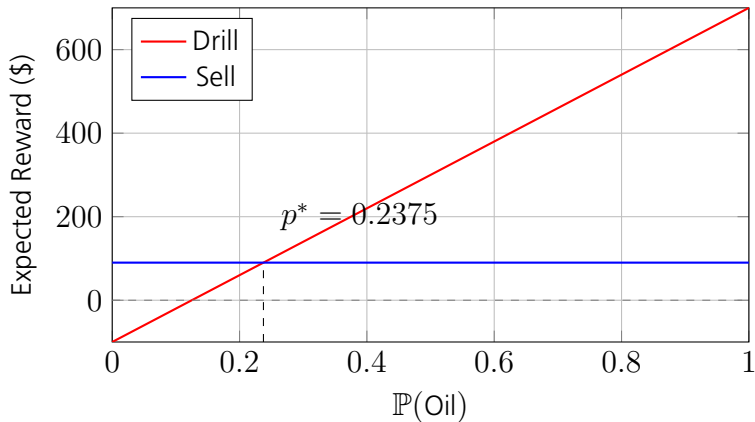
$$800p - 100 = 90$$

$$800p = 190$$

$$p = 0.2375$$

- If $\mathbb{P}(\text{Oil}) > 0.2375$, choose **drill**.
- If $\mathbb{P}(\text{Oil}) < 0.2375$, choose **sell**.

Expected Value vs. Probability of Oil



The two lines cross at $p^* = 0.2375$, the break-even probability.

- The “drill” option is risky but can yield much higher payoff.
- The “sell” option is safe and fixed.
- If you believe there’s at least a 24% chance of oil, drilling is optimal.

St. Petersburg Paradox

- A gamble (at a price $\pi > 0$) is offered to you.
- The gamble pays 2^{n-1} if the **first** heads in tossing a fair coin occurs at the $n \geq 1$ -th toss.
- If you take the gamble the rewards are:

$$r(s, a = \text{play}) = \begin{cases} \$1 - \pi & s = \text{first head in 1 toss} \\ \$2 - \pi & s = \text{first head in 2 tosses} \\ \vdots & \vdots \\ \$2^{n-1} - \pi & s = \text{first head in } n \text{ tosses} \\ \vdots & \vdots \end{cases}$$

1. The Game

- You toss a fair coin until the first head appears.
- Let n be the toss on which the first head occurs.
- The payoff is:

$$\text{payoff}(n) = 2^{n-1} \text{ dollars}$$

- Example:
 - Head on 1st toss ($n = 1$): win $2^0 = 1$
 - Head on 2nd toss ($n = 2$): win $2^1 = 2$
 - Head on 3rd toss ($n = 3$): win $2^2 = 4$
 - ... and so on.

2. Probabilities

- The coin is fair: $\mathbb{P}(\text{Head}) = \frac{1}{2}$.
- To get the first head on toss n :

$$\underbrace{T, T, \dots, T}_{n-1 \text{ tails}}, H$$

- So

$$\mathbb{P}(\text{first head on } n) = \left(\frac{1}{2}\right)^n$$

- (You need $n - 1$ tails and then 1 head.)

3. Expected Reward

- Expected value = sum over all outcomes of

$$(\text{payoff}) \times (\text{probability})$$

- Here that is:

$$\mathbb{E}[\text{payoff}] = \sum_{n=1}^{\infty} (2^{n-1}) \left(\frac{1}{2}\right)^n$$

- Each term = "win 2^{n-1} if first head on n " $\times$ "probability that first head is on n ".

4. Simplifying Each Term

$$2^{n-1} \left(\frac{1}{2} \right)^n = 2^{n-1} \cdot \frac{1}{2^n} = \frac{2^{n-1}}{2^n} = \frac{1}{2}$$

So every term in the sum is $\frac{1}{2}$.

$$\mathbb{E}[\text{payoff}] = \sum_{n=1}^{\infty} \frac{1}{2}$$

5. Divergence

- The sum

$$\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots$$

grows without bound.

- Therefore

$$\mathbb{E}[\text{payoff}] = +\infty$$

- This is the heart of the St. Petersburg paradox:
the *expected* reward is infinite.

6. Price π to Play

- You pay once not every round.
- Expected *net* reward:

$$\mathbb{E}[\text{net}] = \underbrace{\sum_{n=1}^{\infty} 2^{n-1} \left(\frac{1}{2}\right)^n}_{=\infty} - \pi = +\infty$$

- So even after subtracting any finite entry fee π , the expected value is still infinite.

7. Why a Paradox

- Math says: expected value is infinite $\Rightarrow$ pay any finite amount.
- People actually won't pay very much.
- This gap motivates ideas like:
 - diminishing marginal utility of money
 - risk aversion
 - bounded payoffs

Risk Aversion

- Suppose you must choose between the following two gambles:

$$r(s, a_1) = \begin{cases} \$20 & s = s_1 \\ 0 & s = s_2 \\ 0 & s = s_3 \end{cases} \quad r(s, a_2) = \begin{cases} 0 & s = s_1 \\ \$20000 & s = s_2 \\ 0 & s = s_3 \end{cases}$$

with $\mathbb{P}(s_1) = \frac{1}{2}$, $\mathbb{P}(s_2) = \frac{1}{2000}$, $\mathbb{P}(s_3) = 1 - \mathbb{P}(s_1) - \mathbb{P}(s_2)$.

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- $\mathbb{E}_{s \sim \mathbb{P}}[r(s, a_1)] = \mathbb{E}_{s \sim \mathbb{P}}[r(s, a_2)] = 10$
- The choice of a_1 over a_2 indicates *risk aversion*.

Expected Utility

- We consider a disjoint and exhaustive list of **states of the world** $\{s_1, s_2, \dots, s_n\}$ with probability $p_i := p(s_i) \in [0, 1]$ with $\sum_{i=1}^n p_i = 1$.
- A choice of action $a \in A = \{a_1, \dots, a_m\}$ has an associated random variable

$$r(s, a) = \begin{cases} r(s_1, a) & \text{with probability } p_1 \\ \vdots & \\ r(s_n, a) & \text{with probability } p_n \end{cases}$$

Expected Utility

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Definition:

A structure of **preferences** over action set A is a relation $\succeq$, i.e. $a \succeq a'$ means action $a \in A$ is preferred over $a' \in A$.

For example, $a \succeq a'$, if and only if $\mathbb{E}_{s \sim \mathbb{P}}[r(s, a)] \geq \mathbb{E}_{s \sim \mathbb{P}}[r(s, a')]$.

Expected Utility and Risk Aversion

- Let $U : \mathbb{R} \mapsto \mathbb{R}$ a monotone increasing function or “utility”.
- The **expected utility** associated with action a is

$$\mathbb{E}_{s \sim \mathbb{P}}[U(r(s, a))] \quad := \quad \sum_{i=1}^n p_i U(r(s_i, a))$$

- In “Theory of Games and Economic Behavior” (1944)
Von Neumann-Morgenstern introduce a structure of preferences as follows:

Expected Utility Preferences:

$$a \succeq a' \quad \text{if} \quad \mathbb{E}_{s \sim \mathbb{P}}[U(r(s, a))] \geq \mathbb{E}_{s \sim \mathbb{P}}[U(r(s, a'))]$$

Back to Example II

Introducing Risk Aversion

- Suppose utility of wealth is:

$$U(x) = \sqrt{x} + 100$$

- Concave shape $\Rightarrow$ diminishing marginal utility of money.
- We compare **expected utility**, not expected dollars.

4. Expected Utility Formulation

Let initial wealth = w .

$$EU_{\text{drill}} = pU(w + 700) + (1 - p)U(w - 100)$$

$$EU_{\text{sell}} = U(w + 90)$$

Using $U(x) = \sqrt{x} + 100$:

$$EU_{\text{drill}} = p\sqrt{w + 700} + (1 - p)\sqrt{w - 100} + 100,$$

$$EU_{\text{sell}} = \sqrt{w + 90} + 100.$$

The constants cancel, so compare:

$$p\sqrt{w + 700} + (1 - p)\sqrt{w - 100} \quad \text{vs.} \quad \sqrt{w + 90}.$$

5. Solving for the Indifference Probability

Example: Let initial wealth $w = 1000$.

$$p\sqrt{1700} + (1 - p)\sqrt{900} = \sqrt{1090}$$

Solve for p :

$$p = \frac{\sqrt{1090} - \sqrt{900}}{\sqrt{1700} - \sqrt{900}} \approx 0.268$$

Result

Threshold probability $p^* = 0.268$ (26.8%)

- Higher than 23.75% from the risk-neutral case.
- Drilling must be more likely to succeed to justify the extra risk.

- Risk-averse individuals value certainty over risky payoffs.
- Concave $U(x)$ reduces the attractiveness of large but uncertain gains.
- With $U(x) = \sqrt{x} + 100$:
 - Expected **utility** from drilling $<$ from selling unless $p > 0.268$.
 - Expected **money** threshold (risk-neutral): $p > 0.2375$.
- This gap quantifies **risk aversion**.

Back to Example II

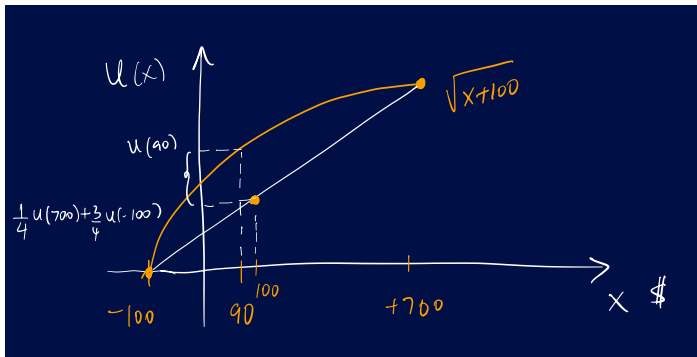


Figure 2: Risk Averse Oil Drilling Decision

Concavity shows diminishing marginal utility (the same \$ gain adds less happiness at higher wealth).

Solution to St. Petersburg Paradox

- With marginally decreasing utility, the expected utility is finite.
- For example, if $U(x) = \sqrt{x}$ we have

$$\begin{aligned}\mathbb{E}[U(r(s, a = \text{play}))] &= \sum_{n=1}^{\infty} \frac{U(2^{n-1} - \pi)}{2^n} \\ &= \sum_{n=1}^{\infty} \frac{(2^{n-1} - \pi)^{\frac{1}{2}}}{2^n} \\ &< \sum_{n=1}^{\infty} \frac{1}{\sqrt{2^{n+1}}} < \infty\end{aligned}$$

1. Paradox Recap

- In the St. Petersburg game:
 - You toss a fair coin until the first head appears.
 - If the first head appears on toss n , you win 2^{n-1} dollars.
- Probability of this outcome:

$$\mathbb{P}(n) = \left(\frac{1}{2}\right)^n$$

- Expected *monetary* value:

$$\mathbb{E}[r] = \sum_{n=1}^{\infty} 2^{n-1} \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} \frac{1}{2} = +\infty$$

- Infinite expectation $\Rightarrow$ paradox: no one would pay an infinite amount to play!

2. Bernoulli's Idea: Diminishing Marginal Utility

- People value *utility* of money, not money itself.
- The utility function $U(x)$ increases with x , but at a decreasing rate.
- Example of a concave (risk-averse) utility:

$$U(x) = \sqrt{x}$$

- Marginal utility:

$$U'(x) = \frac{1}{2\sqrt{x}} \quad \text{decreases as } x \text{ increases.}$$

- Hence, large payoffs add little extra satisfaction.

3. Expected Utility Formulation

If it costs π to play the game, the reward for outcome n is:

$$r_n = 2^{n-1} - \pi$$

The expected utility becomes:

$$\mathbb{E}[U(r)] = \sum_{n=1}^{\infty} \mathbb{P}(n) U(r_n)$$

Substituting $\mathbb{P}(n) = \frac{1}{2^n}$ and $U(x) = \sqrt{x}$:

$$\mathbb{E}[U(r)] = \sum_{n=1}^{\infty} \frac{\sqrt{2^{n-1} - \pi}}{2^n}$$

4. Bounding the Series

For large n , π is negligible compared to 2^{n-1} , so:

$$\sqrt{2^{n-1} - \pi} < \sqrt{2^{n-1}} = 2^{(n-1)/2}$$

Therefore:

$$\frac{\sqrt{2^{n-1} - \pi}}{2^n} < \frac{2^{(n-1)/2}}{2^n}$$

Simplify the right-hand side:

$$\frac{2^{(n-1)/2}}{2^n} = \frac{1}{2^{(n+1)/2}}$$

So the entire series is bounded by:

$$\mathbb{E}[U(r)] < \sum_{n=1}^{\infty} \frac{1}{2^{(n+1)/2}}$$

5. Checking Convergence

- The series

$$\sum_{n=1}^{\infty} \frac{1}{2^{(n+1)/2}}$$

is a geometric series.

- Common ratio:

$$r = \frac{1}{\sqrt{2}} \approx 0.707 < 1$$

- Therefore, it converges:

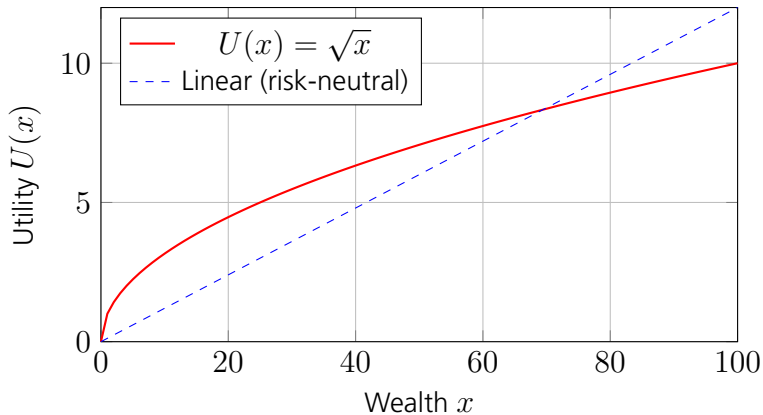
$$\sum_{n=1}^{\infty} \frac{1}{2^{n/2}} = \frac{1}{\sqrt{2} - 1} < \infty$$

- Expected utility is finite!! :)

- With diminishing marginal utility, very large prizes add little to total satisfaction.
- The concave $U(x)$ “compresses” large payoffs.
- Thus:

$$\mathbb{E}[r] = \infty \quad \text{but} \quad \mathbb{E}[U(r)] < \infty$$

- Rational, risk-averse players value the game at a finite amount
→ paradox resolved.



Concavity shows risk aversion: utility increases with x but at a decreasing rate.

Solution to St. Petersburg Paradox, Illustration

Compute approximate expected utilities for different entry fees π (assuming first 10 outcomes):

$$\mathbb{E}[U(r)] \approx \sum_{n=1}^{10} \frac{\sqrt{2^{n-1}} - \pi}{2^n}$$

Entry fee π	Approx. Expected Utility
0	1.41
1	1.38
2	1.36
5	1.31
10	1.26

- Expected utility **declines slowly** as the price increases.
- Beyond a modest π , utility gain from playing no longer compensates cost.
- Hence, a rational player would only pay a few dollars, not an infinite amount.